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# =====
# 1. IMPORT LIBRARIES
# =====

import numpy as np
import pandas as pd

# =====
# 2. LOAD DATASETS
# =====

movies = pd.read_csv('tmdb_5000_movies\\tmdb_5000_movies.csv')
credits = pd.read_csv('tmdb_5000_credits\\tmdb_5000_credits.csv')

# =====
# 3. INITIAL DATASET INSPECTION
# =====

display(movies.shape, credits.shape) # Display the shapes of both DataFrames

display("movies",movies.head(3)) # Display the first 3 rows of the movies DataFrame
display("credits",credits.head(3)) # Display the first 3 rows of the credits DataFrame

credits.head(2)['cast'].values

# =====
# 4. MERGE DATASETS
# =====

# merging both dataframes
movies = movies.merge(credits,on='title') # Merge movies and credits DataFrames on the 'title' column
display("Merged DataFrame",movies.head(3)) # Display the first 3 rows of the merged DataFrame
movies.info()

# =====
# 5. SELECT IMPORTANT COLUMNS
# =====

# imp column selection
movies = movies[['movie_id','title','overview','genres','keywords','cast','crew']] # Extract relevant columns
movies.info()
movies.head(2)

# =====
# 6. HANDLE MISSING VALUES
# =====

# Check for null values and handle them
movies.isnull().sum() # Check for null values
movies.dropna(inplace=True) # Remove rows with null values
movies.isnull().sum() # Verify that no null values remain

# =====
# 7. HANDLE DUPLICATE VALUES
# =====

# Check for duplicate rows and handle them
movies.duplicated().sum() # Check for duplicate rows
movies.drop_duplicates(inplace=True) # Remove duplicate rows
movies.duplicated().sum() # Verify that no duplicate rows remain

# =====
# 8. ACCESS DATA FROM DATAFRAME
# =====

# Accessing specific data in the DataFrame in different ways
movies.iloc[0,3] # Access the 'genres' column of the first row if you know the index of the column.
movies.iloc[0].genres # Access the 'genres' column of the first row if you don't know the index of the column.

movies['genres'][0] # Access the 'genres' column of the first row without using iloc.

# =====
# 9. CONVERT GENRES AND KEYWORDS DATA
# =====

# '[{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"}...]'
# extract only name
# and apply on all dictionaries from original list because original list contains less ammount of dictionaries in 'genres' and 'keywords' columns.
# result: ["Action", "Adventure"...]

import ast

def convert(obj):
    L = []
    for i in ast.literal_eval(obj):
        L.append(i['name'])
    return L

movies['genres'] = movies['genres'].apply(convert) # Convert the 'genres' column to a list of genre names

movies.head(2) # confirm the changes made to the 'genres' column

movies['keywords'] = movies['keywords'].apply(convert) # Convert the 'keywords' column to a list of keyword names

movies.head(2) # confirm the changes made to the 'keywords' column

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# =====
# 10. EXTRACT TOP 3 CAST MEMBERS
# =====

# [{"id": 28, "name": "Action"}, {"id": 12, "name": "Adventure"...}]
# extract only name
# and apply on first 3 dictionaries from original list because original list contains huge ammount of dictionaries in 'cast' column.
# result: ["Action", "Adventure"...]

def convert3(obj):
    L = []
    counter = 0
    for i in ast.literal_eval(obj):
        if counter < 3: # Limit to the first 3 items
            L.append(i['name'])
            counter += 1
        else:
            break
    return L

movies['cast'] = movies['cast'].apply(convert3) # Convert the 'cast' column to a list of the first 3 cast member names

movies.head(2) # confirm the changes made to the 'cast' column

# =====
# 11. EXTRACT DIRECTOR FROM CREW
# =====

# [{"credit_id": "52fe48009251416c750aca23", "department": "Directing",...}]
# extract only department

def fetch_director(obj):
    L = []
    for i in ast.literal_eval(obj):
        if i['job'] == 'Director':
            L.append(i['name'])
            break # Stop after finding the first director
    return L

movies['crew'] = movies['crew'].apply(fetch_director) # Convert the 'crew' column to a list containing the director's name

movies.head(2) # confirm the changes made to the 'crew' column

# =====
# 12. CONVERT OVERVIEW INTO A LIST OF WORDS
# =====

movies.iloc[0].overview # Access the 'overview' column of the first row we found it is a string and we will convert it to a list of words.

movies['overview'] = movies['overview'].apply(lambda x:x.split()) # Convert the 'overview' column to a list of words

movies.head(2) # confirm the changes made to the 'overview' column

# =====
# 13. REMOVE SPACES FROM TEXT
# =====

movies['genres'] = movies['genres'].apply(lambda x:[i.replace(" ","") for i in x]) # Remove spaces from genre names
movies['keywords'] = movies['keywords'].apply(lambda x:[i.replace(" ","") for i in x]) # Remove spaces from keyword names
movies['cast'] = movies['cast'].apply(lambda x:[i.replace(" ","") for i in x]) # Remove spaces from cast names
movies['crew'] = movies['crew'].apply(lambda x:[i.replace(" ","") for i in x]) # Remove spaces from crew names

movies.head(2) # confirm the changes made to the 'genres,keywords,cast,crew' column

# =====
# 14. CREATE TAGS COLUMN
# =====

movies['tags'] = movies['overview'] + movies['genres'] + movies['keywords'] + movies['cast'] + movies['crew'] # Combine all relevant columns into a single 'tag'

movies.head(2) # confirm the changes made to the 'tags' column

# =====
# 15. CREATE FINAL DATAFRAME
# =====

new_df = movies[['movie_id','title','tags']] # Create a new DataFrame with only the 'movie_id', 'title', and 'tags' columns

new_df.head(2) # Confirm the changes made to the new DataFrame

# =====
# 16. CONVERT TAGS LIST INTO STRING
# =====

new_df['tags'] = new_df['tags'].apply(lambda x: " ".join(x)) # Convert the 'tags' column from a list of words to a single string of words

new_df.head(2) # Confirm the changes made to the new DataFrame

# =====
# 17. CONVERT TAGS TO LOWERCASE
# =====

new_df['tags'] = new_df['tags'].apply(lambda x:x.lower()) # Convert the 'tags' column to lowercase for uniformity

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new_df.head(2) # Confirm the changes made to the new DataFrame

# =====
# 18. APPLY STEMMING
# =====

# ['loved', 'love', 'loves', 'loving']
# after applying stemming to reduce words to their root form. we will get only 'love' as a feature.
# ['love']

import nltk

from nltk.stem.porter import PorterStemmer
ps = PorterStemmer()

def stem(text):
    y = []
    for i in text.split():
        y.append(ps.stem(i)) # Apply stemming to each word in the text
    return " ".join(y) # Join the stemmed words back into a single string

new_df['tags'] = new_df['tags'].apply(stem) # Apply the stemming function to the 'tags' column

# =====
# 19. Vactorisation
# =====

from sklearn.feature_extraction.text import CountVectorizer

cv = CountVectorizer(max_features=5000, stop_words='english') # Create a CountVectorizer object to convert text data into numerical features
vector = cv.fit_transform(new_df['tags']).toarray() # Fit and transform the 'tags' column to create a feature matrix

vector # Display the shape of the feature matrix

list(cv.get_feature_names_out()) # Display the feature names

# =====
# 20. COSINE SIMILIARITY
# =====

from sklearn.metrics.pairwise import cosine_similarity

similarity = cosine_similarity(vector) # Compute the cosine similarity between all pairs of movies based on their feature vectors

# =====
# 21. SORTING THE SIMILIARITY AND RECOMMENDATION FUNCTION
# =====

sorted(list(enumerate(similarity[0])), reverse=True, key=lambda x: x[1])[1:6] # Sort the movies based on similarity to the first movie and display the top 5 simi

def recommend(movie):
    movie_index = new_df[new_df['title'] == movie].index[0] # Get the index of the movie in the DataFrame
    distances = similarity[movie_index] # Get the similarity scores for the movie
    movies_list = sorted(list(enumerate(distances)), reverse=True, key=lambda x: x[1])[1:6] # Sort the movies based on similarity and get the top 5
    for i in movies_list:
        print(new_df.iloc[i[0]].title) # Print the titles of the recommended movies

# =====
# 22. TEST THE MODEL
# =====

recommend('Batman Begins') # Call the recommend function to get movie recommendations based on 'Batman Begins'

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